Response to Reviewers — Second Cut

Team 3 · *Servo Characterization for the SOFIA Modular Kinetic Surface*

Acknowledgements

The authors thank the reviewers for their careful evaluation of the first-cut manuscript. Their comments helped clarify the research scope, experimental methodology, and electrical compatibility. All comments were considered; the responses below follow the keys of the review report (R1.1, R1.2, . . .). Line, section, figure, table, and equation numbers refer to the revised manuscript that accompanies this letter.

Main changes. (i) Refined problem statement and objectives (Section I). (ii) Expanded characterization methodology (Section II). (iii) Documented RS-485 voltage-level adaptation (Section II-E).

Summary of changes (observation–response–change matrix)

23 observations: 12 addressed, 11 partially addressed, 0 not addressed (justified).

Key	Status	Location in the revised paper
R1.1	**[Addressed]**	Title, author block, and running header
R1.2	**[Addressed]**	line 165, line 183, line 186, line 189, and line 192
R1.3	**[Addressed]**	Table II
R1.4	**[Addressed]**	Tables I, II, and III; Fig. 1; Algorithm 1
R1.5	**[Partially addressed]**	Sections IV, V, and VI
R1.6	**[Partially addressed]**	Table VIII and manuscript source
R2.1	**[Addressed]**	line 1 and line 23
R2.2	**[Addressed]**	Section II, line 134, and line 141
R2.3	**[Partially addressed]**	line 183, line 186, line 189, and line 192
R2.4	**[Addressed]**	line 36 and line 124
R3.1	**[Addressed]**	Fig. 1 and line 170
R3.2	**[Partially addressed]**	line 141, line 153, and (1)
R3.3	**[Addressed]**	line 241, Table VI, and (1)
R3.4	**[Addressed]**	Section II-E and line 208
R3.5	**[Addressed]**	Tables IV, V, and VI
R3.6	**[Partially addressed]**	Table VIII and line 222
R3.7	**[Partially addressed]**	Tables IX, X, XI, and XII
R3.8	**[Partially addressed]**	Algorithm 1 and line 260
R4.1	**[Partially addressed]**	line 124, line 141, and (1)
R4.2	**[Addressed]**	Section I and lines 49–60
R4.3	**[Partially addressed]**	line 113 and line 97
R4.4	**[Partially addressed]**	line 27 and Fig. 2
R4.5	**[Partially addressed]**	Table II

Reviewer 1 — *Format, LaTeX and template compliance*

R1.1 [Addressed]

Reviewer comment: El título es “LAB-DST SOFIA O26”, que no describe el trabajo. No hay afiliación en `\thanks`. El apellido del profesor aparece como “Gonzales Jimenez”, y el encabezado dice “Technical Datasheet of SOFIA Demonstrator Module”.

Response: We revised the manuscript title, author information, institutional affiliation, and running header to reflect the servo-characterization focus and comply with the IEEE journal format.

Changes in the revised paper: Title, author block, and running header

R1.2 [Addressed]

Reviewer comment: Las tablas de requisitos, de la ESP32-C3, del PCA9685, del MG90S y de verificación no tienen `\label` y no se citan en el texto; aparecen una tras otra sin párrafo que las introduzca. Las fuentes se colocan como líneas sueltas ("Source: ESP32-C3 Datasheet, Espressif Systems.", Línea 105).

Response: We added labels and in-text references for the requirements and technical-verification tables, introduced each table in the surrounding text, and moved the component-source citations into the corresponding captions.

Changes in the revised paper: line 165, line 183, line 186, line 189, and line 192

R1.3 [Addressed]

Reviewer comment: En la tabla de benchmarking el texto de las columnas se encima (p. ej. "Continuous (non-discrete)" invade la columna Modularity).

Response: We reformatted the benchmarking table using column-width-constrained tabularx columns and smaller text to reduce column overflow.

Changes in the revised paper: Table II

R1.4 [Addressed]

Reviewer comment: Todos los flotantes usan [H] (paquete float).

Response: The figures, tables, and algorithm now use IEEE-style float placement options instead of forced [H] placement.

Changes in the revised paper: Tables I, II, and III; Fig. 1; Algorithm 1

R1.5 [Partially addressed]

Reviewer comment: Faltan las secciones Results, Discussion y Conclusions (al menos como encabezados) y todas las secciones de cierre.

Response: We added the Results, Discussion, and Conclusions headings, including preliminary experimental subsections. The experimental findings and required closing statements are not yet included.

Changes in the revised paper: Sections IV, V, and VI

R1.6 [Partially addressed]

Reviewer comment: Al final del archivo hay una tabla comentada con valores de ejemplo ("Manufacturer Y, City, Country", "Continuous-rotation servo, rated torque τ at V V"), copiada del ejemplo de la clase de BOM. En figures/ hay cinco imágenes (nop (1..5).png) que no se usan, con espacios en el nombre.

Response: The unused commented BOM example is absent from the revised manuscript. The separate figures directory was not available for verification, so removal of unused images remains unconfirmed.

Changes in the revised paper: Table VIII and manuscript source

Reviewer 2 — *Writing and structure*

R2.1 [Addressed]

Reviewer comment: El abstract y las palabras clave eran el texto guía de la plantilla; en esta versión quedaron vacíos.

Response: We replaced the template text with a preliminary research abstract and six IEEE keywords, focusing on servo-motor dynamic-response characterization in one SOFIA module.

Changes in the revised paper: line 1 and line 23

R2.2 [Addressed]

Reviewer comment: Materials and Methods se presenta como otro documento: “The objective of this document is to verify the compatibility of the selected components” y “The document structure follows the LabDST documentation requirements for technical datasheets”. Es la ficha técnica de la clase de hardware insertada tal cual.

Response: We rewrote the opening of Materials and Methods as part of the SOFIA research article, presenting the system, its experimental scope, and the proposed stages of hardware configuration, measurement, and response analysis instead of a standalone technical datasheet.

Changes in the revised paper: Section II, line 134, and line 141

R2.3 [Partially addressed]

Reviewer comment: Salvo un párrafo final, Materials and Methods está formada solo por tablas; las subsecciones A–F no tienen texto que explique qué muestra cada tabla ni qué se concluye.

Response: Explanatory paragraphs now introduce the main specification and verification tables in Materials and Methods. Further interpretation of individual compliance findings is still needed.

Changes in the revised paper: line 183, line 186, line 189, and line 192

R2.4 [Addressed]

Reviewer comment: La introducción afirma que “The SOFIA system integrates three main capabilities: automatic configuration and communication of modules through a Modbus network, characterization of the mechanical time constant of the servomotors...” como si ya existieran, cuando son la propuesta del equipo y no se desarrollan en el texto.

Response: We revised the Introduction to distinguish proposed research from implemented system functions. The scope is now focused on servomotor dynamic-response characterization, while automatic Modbus addressing is not claimed as a research contribution.

Changes in the revised paper: line 36 and line 124

Reviewer 3 — *Technical content*

R3.1 [Addressed]

Reviewer comment: No hay arquitectura preliminar del sistema en figura: cómo se conectan la cámara y el PC, la ESP32-C3, los dos PCA9685, los 32 servos, los transceptores RS-485 y la alimentación.

Response: We added a preliminary SOFIA system architecture diagram and linked its description to Fig. 1, covering the host, embedded controller, PWM drivers, and servomotor connections.

Changes in the revised paper: Fig. 1 and line 170

R3.2 [Partially addressed]

Reviewer comment: No hay matriz de riesgos, micropruebas ni protocolo experimental para las dos propuestas del diferenciador: el tiempo de configuración de red y la constante de máquina de los servos.

Response: We narrowed the experimental scope to servomotor characterization and described the proposed fixture, measured variables, and first-order response model. A complete risk matrix, detailed micro-test protocol, and experimental validation remain pending; automatic Modbus addressing is outside the revised scope.

Changes in the revised paper: line 141, line 153, and (1)

R3.3 [Addressed]

Reviewer comment: La “constante de máquina” o “mechanical time constant” del servo no se define: no hay modelo, variables ni ecuaciones. El diferenciador promete compararla contra el “valor teórico” del MG90S, que la hoja de datos no reporta.

Response: We introduced a first-order position-response model defining θ_0, θ_f, and τ_m. The MG90S datasheet provides no-load angular-speed specifications, not a manufacturer-reported mechanical time constant; experimental estimation remains necessary.

Changes in the revised paper: line 241, Table VI, and (1)

R3.4 [Addressed]

Reviewer comment: El BOM incluye 16 transceptores “MAX13487E TTL485-V2.0”. Según su hoja de datos, ese transceptor opera con alimentación de 5 V (verifíquenlo), y su salida de recepción quedaría a niveles de 5 V en una entrada de la ESP32-C3, que no tolera 5 V, como ustedes mismos anotan en la Tabla IV.

Response: We added the subsection “RS-485 Logic-Level Compatibility” to Materials and Methods, documenting the measured logic-level incompatibility and the resistive voltage divider incorporated into the revised PCB schematic. Validation of the assembled PCB remains pending.

Changes in the revised paper: Section II-E and line 208

R3.5 [Addressed]

Reviewer comment: Las fichas técnicas tienen solo parámetro, valor y verificación. No indican la versión ni la página de la hoja de datos, ni justifican el parámetro frente al requerimiento. El par del MG90S aparece como “1.8–2.2 kgf cm” sin decir a qué voltaje corresponde cada valor.

Response: We expanded the ESP32-C3, PCA9685, and MG90S specification tables with source information and parameter justifications, and specified the MG90S torque values at their corresponding supply voltages.

Changes in the revised paper: Tables IV, V, and VI

R3.6 [Partially addressed]

Reviewer comment: El BOM no distingue la cantidad por unidad de la del sistema. Lista 1 ESP32-C3 y 2 PCA9685 pero 16 transceptores RS-485, sin explicar la arquitectura que eso implica. Usa “3D Printed” como fabricante, contiene elementos poco claros (“Pro3” de Tower Pro, “Axis 1 9mm x 9mm”, “Inser M3x5x4”) y lista conectores JST-PH, cuando la arquitectura de SOFIA usa JST-VH. Tampoco tiene descripción técnica ni localidad.

Response: We revised the BOM to distinguish quantities per unit and per system and to include technical descriptions, manufacturers, and origins. The listed transceiver is identified as controller-side; a complete count for future distributed modules and some component descriptions still require clarification.

Changes in the revised paper: Table VIII and line 222

R3.7 [Partially addressed]

Reviewer comment: La matriz de alternativas tiene un solo criterio por función, sin ponderación ni calificación.

Response: We expanded the alternative comparison into separate tables for actuation, PWM generation, processing, and communication. Weighted criteria and numerical ratings have not yet been included.

Changes in the revised paper: Tables IX, X, XI, and XII

R3.8 [Partially addressed]

Reviewer comment: El algoritmo usa r_0 para la posición del módulo y $\hat{v}$ para la dirección, en lugar de la notación del curso ($r_{0,2}$, $\hat{r}_{2,p}$). Además escribe atan2, asin y Clamp en itálica matemática, sin `\operatorname` ni `\Call`.

Response: We updated the pseudocode to use the course coordinate notation, standard mathematical operators, and the ClampAngle call, and identified the smoothing update as an EMA. A specific numerical smoothing-factor setting is not yet reported.

Changes in the revised paper: Algorithm 1 and line 260

Reviewer 4 — *D-Lab and state of the art*

R4.1 [Partially addressed]

Reviewer comment: El diferenciador propone asignación automática de direcciones Modbus y caracterización de la constante de máquina de cada servo. El artículo menciona ambas en la introducción, pero se enfoca en justificar la arquitectura electrónica (Línea 83) y no desarrolla ninguna de las dos.

Response: We narrowed the research contribution to experimental servomotor characterization and documented a proposed first-order model and methodology. Automatic Modbus addressing is outside the revised scope; experimental time-constant identification has not yet been completed.

Changes in the revised paper: line 124, line 141, and (1)

R4.2 [Addressed]

Reviewer comment: No hay planteamiento formal del problema, pregunta de investigación ni objetivos.

Response: We added a problem statement, research question, and three objectives to the Introduction, addressing individual servomotor characterization and the proposed evaluation of actuator synchronization.

Changes in the revised paper: Section I and lines 49–60

"Standard low-cost micro-servos exhibit manufacturing tolerances that result in non-uniform actuation speeds and transient response times across different units. In a 16-unit, 32-actuator modular kinetic surface like SOFIA, driving all servos with uniform control signals creates dynamic motion desynchronization and mechanical stress during normal use. To address this limitation, this work investigates the following research question: Can experimental estimation of individual servo mechanical time constants (τ_m) and their integration into a trajectory controller improve dynamic positional synchronization across a 32-actuator kinetic surface? This encompasses three primary objectives: (1) designing a PCB integrating an ESP32-C3 microcontroller, dual PCA9685 drivers, and RS-485 interfaces; (2) developing an automated micro-test protocol to capture step-response and extract individual time constants; and (3) integrating individual compensation parameters into the surface tracking controller to equalize dynamic settling times across all actuators."

R4.3 [Partially addressed]

Reviewer comment: El demostrador se describe por su composición (16 unidades, 32 servos, una ESP32-C3 y dos PCA9685), pero no hay despiece funcional, diagrama de contexto, árbol de problemas ni bitácora de observaciones. Las oportunidades se derivan de la literatura, no de observar el demostrador.

Response: We expanded the description of SOFIA and the functional opportunities identified in the Introduction. A functional breakdown matrix, an observation log, and an evidence-based Discover analysis of the physical demonstrator are still pending.

Changes in the revised paper: line 113 and line 97

R4.4 [Partially addressed]

Reviewer comment: Se describe un arreglo "two-dimensional 4×4"; la simulación muestra columnas escalonadas.

Response: We replaced the earlier 4-by-4 wording with a two-dimensional honeycomb description and included a simulation figure. The precise stepped arrangement and center-to-center dimensions still require verification against the Sofia003 CAD model.

Changes in the revised paper: line 27 and Fig. 2

R4.5 [Partially addressed]

Reviewer comment: El benchmarking usa criterios cualitativos, y la fila de SOFIA se califica "High" en modularidad sin distinguir entre meta de diseño y resultado medido.

Response: We explicitly identify the SOFIA benchmarking entry as proposed rather than measured. However, quantitative benchmarking criteria and measured values have not yet been added.

Changes in the revised paper: Table II